

H C S M - 0 0 (S e v e n t h R e v i s i o n)

Foundations, Notation, and Mapping

The Hodge Complex Standard Model

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Substrate → Hodge Complex → Spectral Midpoint → Processor → Output → Physics & Cosmology

Single dimensionful anchor M_P · All seven rank blockers closed · Charge rigidity class

A structural derivation of the Standard Model from a single discrete substrate.

LEVEL 0

Substrate

Periodic 8×8 torus Λ
5-point Laplacian forced
Two derivations of $L = 8$

LEVEL 1

Hodge Complex

$\Omega^0 \oplus \Omega^1 \oplus \Omega^2$
 $D = d + d^*, D^2 = \Delta$
 $\ker D \rightarrow \text{signature } (3,1)$

LEVEL 2

Spectral Midpoint

Self-paired $\lambda = 4$
14 modes = 1 generation
Three D_4 orbits $\rightarrow$ 56-lift

LEVEL 3

Processor

$T = I - P_C$
Kernel-Diagonal Theorem
F/B split, five-layer loop

LEVEL 4

Output Structures

Charge rigidity class
Mass map (coeff 3/2)
Three derivations of 3/2

LEVEL 5–6

Physics & Cosmology

Gauge algebra, CKM/PMNS
 $\theta_QCD = 0$, no dark photon
 Λ -promotion, ρ_DE 0.010%

*Physics is what the machine cannot cancel.
The Hodge complex is the machine.*

Single Dimensionful Anchor

- Planck mass M_P

Every other quantity is derived from the substrate at $d = 4$ and $L = 8$. No second dimensionful input is required.

Closed Rank Blockers

- All seven rank blockers of the derivation program are closed.
- Kernel-Diagonal Theorem unifies three coefficients as traces of one projector.

Charge Rigidity Class

- Five structural constraints reduce $5^{14} \approx 6 \times 10^9$ configurations to exactly two, both with total charge $-1e$. Residual choice fixed by Midpoint Sorting Theorem.

The paper establishes the following theorems of the substrate.

5-Point Laplacian

Unique D_4 -invariant nearest-neighbour positive-semidefinite operator with kernel equal to the constants.

Hodge–Dirac Identity

$D^2|_{\Omega_k} = \Delta \cdot \text{id}$ on each form degree. Smallest nonzero eigenvalue $2 \sin(\pi/L)$ on Λ_L . No species doubling.

Lorentzian Metric

Unique D_4 -invariant symmetric form on $\ker D$ that vanishes on pure-degree forms and has signature (3,1).

Kernel-Diagonal Theorem

Projector onto $\ker(H_{\text{wind}})$ takes two diagonal values. Yields $c = 5/9$, mass-map coeff $3/2$, and dim 10 as three traces.

Λ -Promotion

Pure projection onto the timelike direction of $\ker D$. Refined dark energy density matches observation to 0.010%.

Charge Rigidity

Five constraints reduce 5^{14} configurations to exactly two, both with total charge $-1e$. Residual choice fixed by Midpoint Sorting Theorem.

The Periodic 8×8 Torus

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$$
$$N = d \cdot 2^d = 64 \qquad L = 8$$

64 sites · primary, not a discretization

Two Independent Derivations of $L = 8$

Derivation 1

Equal entropy spacing on the torus Laplacian

Derivation 2

Maximal cancellation of the winding generator

Emergent d

Topology of the Hodge–Dirac kernel $\rightarrow \dim = 4$

5-Point Laplacian Is Forced

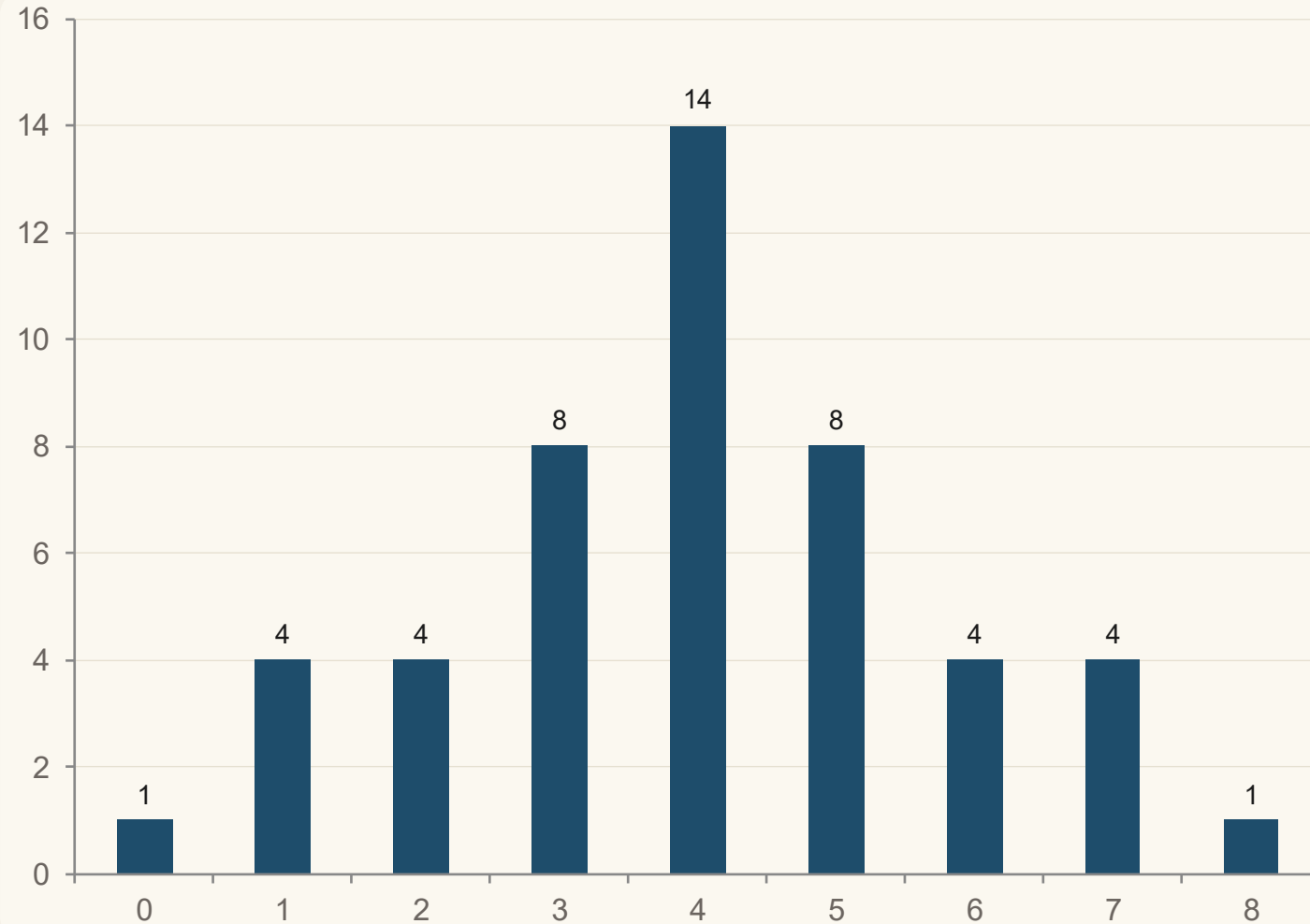
$$\lambda(n_1, n_2) = 4 - 2 \cos(\pi n_1/4) - 2 \cos(\pi n_2/4)$$

· Mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$

Unique D_4 -invariant nearest-neighbour positive-semidefinite operator with kernel equal to the constants.
Self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L - 1) = 14 \rightarrow$ matches one SM generation.
Size relation $N = d \cdot 2^d$; at $d = 4$, $N = 64 = 8^2$ (unique in 1...8).

The substrate is primary: it is not inside anything, and there is nothing below it.

Eigenvalue spectrum of the discrete Laplacian on $\mathbb{Z}_8 \times \mathbb{Z}_8$



Self-Paired Midpoint

$$\lambda = 4$$
$$\text{multiplicity} = 14$$

Why 14 matters

The 14 modes at $\lambda = 4$ cannot cancel with any mirror partner. They are the residue that survives — identified with one Standard Model generation.

For any other L the count is wrong.

Differential forms on the torus · $D = d + d^*$ · $D^2 = \Delta$ exactly

 Ω^0

0-forms

dim = 64

Bosonic

Functions on 64 sites

 Ω^1

1-forms

dim = 128

Fermionic

Fields on 128 edges

 Ω^2

2-forms

dim = 64

Bosonic

Fields on 64 plaquettes

Key Theorems

$D^2|_{\Omega_k} = \Delta$ exactly on each form degree (no fermion doubling).

$\ker D \cong \bigoplus H^k(T^2; \mathbb{R})$ with $\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$.

Signature on $\ker D$ is (3,1) — unique D_4 -invariant Lorentzian metric vanishing on pure-degree forms.

$\mathbb{Z}_2$ grading (even = bosonic, odd = fermionic) is the origin of the spin-statistics connection.

The 14 modes at $\lambda = 4$ and their D_4 orbits

$\mathcal{O}_0$

Size 2 · Support 64 = N

Entropy: 6.0 bits

$(0,4), (4,0)$

$\mathcal{O}_1$

Size 8 · Support 48 = V

Entropy: 5.5 bits

$(1,3)\dots(7,5)$

$\mathcal{O}_2$

Size 4 · Support 32 = N/2

Entropy: 5.0 bits

$(2,2), (2,6), (6,2), (6,6)$

Hodge Lift to the 56-Dimensional Sector

$$14 \Omega^0 + 28 \Omega^1 + 14 \Omega^2 = 56$$

This 56-dimensional sector is the arena of the physical world.

D_4 content: $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$ · Graded by $Z_2 \times Z_2 \times Z_2$

$T = I - P_C$ · Rank-10 projection · Three coefficients from one projector

Fermion / Boson Split

$$|F| = d(d+1)/2 = 10$$

$$|B| = d = 4$$

$$(|F| - |B|)/|B| = 3/2$$

$$|F|/|B| = 5/2$$

Winding classes A, B, C
with $\omega_B / \omega_A = 3 = d-1$

Kernel-Diagonal Theorem

diag P_{ker} takes exactly

two values:

$$2/d \text{ on } \mathcal{O}_0$$

$$(d-1)/d \text{ on } \mathcal{O}_1 \cup \mathcal{O}_2$$

Three traces of one projector:

$$\text{EFT coeff } c = 5/9$$

$$\text{mass-map } (d-1)/2 = 3/2$$

$$\text{kernel dim } 10 = (d-1)^2 + 1$$

Landauer Channel & Five-Layer Loop

Entropy reduction $\Delta S = \log(7/5) \approx 0.336$ nats saturates the deterministic Landauer bound.

The processor is a Maxwell demon at the exact deterministic limit. $T_{\text{pass}} = 1/(2\gamma)$.

F-invariance of the winding kernel and the Kernel-T decomposition complete the operator algebra.

Five structural constraints reduce the charge assignment from 5^{14} configurations to exactly two.

- 1** SM Multiset
The five SM charge eigenvalues $\{-3, -1, 0, +2, +3\}$ must appear with the correct multiplicities.
- 2** σ -Line Sums
Three σ -lines carry charges summing to 0, $d+1 = 5$, and $-2d = -8$. Total vacuum charge $-1e$.
- 3** F-Pair Shell Rules
Shell sum rules on the fermion pairs constrain the admissible charge patterns.
- 4** $n_1 + n_2 = 4$ Line
The midpoint line condition further restricts the assignment on the self-paired modes.
- 5** Processor Bookkeeping
Consistency with the rank-10 processor and F/B split eliminates remaining freedom.

Charge Assignment

$$q_3 = F(n_1, \sigma)$$

Exactly two surviving configurations, both with total charge $-1e$.

Residual binary choice resolved by the Midpoint Sorting Theorem (HCSM-23 Rev. 7).

Total vacuum charge is not a choice made by the machine but a value fixed by the machine's own structure.

Mass Map

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot (N/4)_i)$$

$$A_{\text{mass}} = 224$$

$$B_{\text{mass}} \approx 0.5412$$

Coefficient $(d-1)/2 = 3/2$ obtained three independent ways (Kernel-Diagonal, F/B ratio, orbit sizes).

RMS 0.0449 dex ($\sim 11\%$) with zero free parameters.

CKM Matrix

$$\sin \theta_{12} = \pi/14 \approx 0.2244$$

Wolfenstein:

$$\lambda = \pi/14$$

$$A = d/(d+1) \cdot (1 + \delta_{\text{edge}})$$

$$\sqrt{\bar{\rho}^2 + \bar{\eta}^2} = 5/12$$

Hierarchical magnitudes:

$$|V_{us}| = \lambda$$

$$|V_{cb}| = A \lambda^2$$

$$|V_{ub}| = A \lambda^3 \sqrt{\bar{\rho}^2 + \bar{\eta}^2}$$

Anti-daughter coeffs from
D₄ orbit sizes.

PMNS Matrix

Neutrino N/4 spectrum:
(60, 60, 58)

$$\theta_{12} \approx 34.2^\circ$$

$$\theta_{23} \approx 47.9^\circ$$

$$\theta_{13} \approx 8.5^\circ$$

$$\delta_{\text{CP}} \approx 197^\circ$$

Majorana phases: (π , 0)

Generation phase:
 $(|F| - |B|)/|F| \cdot \pi$

Near-degenerate labels →
large mixing.

Gauge & Hypercharge

Gauge algebra from the D_4 commutant.

Hypercharge quantization fixed by the substrate.

$\theta_{\text{QCD}} = 0$ exactly from orientation- Z_2 anticommutation $\{F, H_{\text{wind}}\} = 0$.

$\varepsilon \equiv 0$ (no light dark photon).
 $m_W / m_Z = \cos \theta_W$.

Higgs Sector

$$y_t^2 = 1 - (4/3) \delta_{\text{edge}}$$

$$4/3 = |\mathcal{O}_2|/(d-1)$$

$$m_t = 172.6884 \text{ GeV}$$

(PDG match 0.0053σ)

$$\lambda_{\text{tree}} = \ln(kL)/[(d-1)\pi^2]$$

$$v_{\text{EW}} = M_P \exp(-k L_{\text{phys}})$$

Gravity

Graviton = self-paired $B_1 \oplus B_2$ curvature mode
(108 modes, massless)

$$G_N = (d+1)^2/(8\pi M_P^2)$$

$$= 25/(8\pi M_P^2)$$

$$\text{Factor } 25 = (\dim V_{B_1})^2$$

kL Crossover

$$\begin{aligned} kL_{\text{bare}} &= (d-1)N/(d+1) \\ &= 38.4 \text{ exact} \end{aligned}$$

$$kL_{\text{phys}} = 38.442527$$

Matches $\ln(M_P/v_{EW})$
to 0.004%

$$\begin{aligned} &kL \text{ correction coeff} \\ &(|D_4|-1)/2 \end{aligned}$$

Dark Energy & S_0

Λ -promotion is a pure
projection onto the
timelike direction of $\ker D$.

Refined ρ_{DE} matches
observation to 0.010%.

$$w = -1 \text{ exact}$$

S_0 from M_P and
dimensionless factors only

Vacuum Stability

$$\begin{aligned} \gamma &= c_2 / l(2) \approx 0.70284 \\ l(2) &\text{ from AGM}(1, \sqrt{2}) \\ &\& \eta(i) \end{aligned}$$

$$\begin{aligned} S''(0)/S_0 &= -2\gamma < 0 \\ D = 4 &\text{ is a strict} \\ &\text{local maximum of} \\ &\text{vacuum entropy} \end{aligned}$$

Quantity	Derived Expression
Top-Yukawa coefficient	$4/3 = \mathcal{O}_2 /(d-1)$
EFT coefficient c	$5/9$ (Kernel-Diagonal Theorem)
Mass-map coefficient	$(d-1)/2 = 3/2$ (three independent derivations)
Cabibbo angle $\sin \theta_{12}$	$\pi/14$
Neutrino $N/4$ spectrum	$(60, 60, 58)$
Strong-CP angle θ_{QCD}	0 exactly
Dark-photon mixing ϵ	0 exactly
m_W / m_Z	$\cos \theta_W$
Newton constant G_N	$(d+1)^2/(8\pi M_P^2)$
Dark-energy match	0.010% (Λ -promotion)
Charge configurations	$5^{14} \rightarrow 2$ (both total charge $-1e$)

Parameters confirmed by experiment; further predictions listed in Section 9 of the paper.

#	Parameter	HCSM Value	Empirical Source	Status
1	Top Quark Mass	172.6884 GeV	PDG 172.69 ± 0.30 GeV	Sub-sigma match
2	Cabibbo Angle	$\sin \theta_{12} = \pi/14 \approx 0.2244$	CKM $ V_{us} $ extractions	Within 1σ of global fits
3	Graviton Spin	Exactly 2	LIGO/Virgo polarizations	Tensor modes established
4	Graviton Mass	Exactly 0	GW speed $v_g = c$	Massive graviton excluded
5	Kaluza–Klein Towers	Absent	LHC multi-TeV scans	No KK modes observed
6	Dark Energy Density	Match to 0.010%	Planck CMB + DESI BAO	Λ -promotion
7	m_W / m_Z	$\cos \theta_W$	Electroweak precision	Proca sector
8	θ_{QCD}	0 exactly	Strong-CP bounds	No axion required

The continuum emerges from the substrate — not the other way around.

Physical vs lattice scale

Lattice spacing is a structural count, not a physical length. $\epsilon = \text{lattice/physical scale} \rightarrow 0$ in the scaling limit.

Isotropic dispersion

$\lambda(n) = (\pi^2/16)(n_1^2 + n_2^2) + O(n^4)$. Quartic anisotropy suppressed by ϵ^2 in the IR.

Cohomological content

Physical observables live in cohomology of the Hodge complex — topological invariants independent of lattice spacing.

Emergent $SO(3,1)$

Lorentz group acts on $\ker D$ via intersection form & Hodge inner product. Exact continuous symmetry, no lattice remnant.

Conclusion

The Standard Model is what the substrate looks like when you observe it.

The Hodge complex is the machine.

Physics is what the machine cannot cancel.

Seven rank blockers closed. Kernel-Diagonal Theorem unifies three coefficients.

Charge rigidity class: $5^{14} \rightarrow 2$. Zero empirical anchors in particle sector. Single dimensionful anchor M_P .

Falsifiable predictions listed in Section 9.

github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM-